

MedPhys DICOM Toolkit: Comprehensive User & Quality Assurance Manual

Version: 1.0

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Date: October 2026

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1. Introduction & Clinical Scope

The **MedPhys DICOM Toolkit** is an advanced, lightweight, open-source web application engineered for medical physicists, clinical researchers, and imaging technologists. Operating entirely within a local browser session, it ensures zero data transmission, guaranteeing absolute compliance with global data privacy frameworks (HIPAA, GDPR).

This toolkit provides an integrated pipeline for:

- Secure clinical data de-identification.
- Rigorous Quality Assurance (QA) and Quality Control (QC) metric extraction.
- Automated dosimetric auditing for the establishment of Diagnostic Reference Levels (DRLs).

2. Module 1: Batch DICOM Anonymizer

The Anonymizer module facilitates the bulk de-identification of DICOM archives while preserving the essential technical, geometric, and dosimetric metadata required for physical analysis.

Features & Compliance

- **Base ID Prefixing:** Users can assign custom prefixes (e.g., ANON_PATIENT_001).
- **Extended PHI Scrubbing:** Aligned with the **DICOM Standard PS3.15 Annex E** (Basic Application Level Confidentiality Profile). When enabled, this feature redacts referring physician names, operator names, accession numbers, and internal study IDs.
- **Date Removal:** Eliminates PatientBirthDate and StudyDate to prevent cross-referencing temporal data.
- **Audit Trail Generation:** Automatically compiles a .csv compliance report mapping original files to their newly assigned pseudonyms for institutional record-keeping.

3. Module 2: Inspector & 3D Volumetric Viewer

This module serves as the primary analytical engine. It dynamically adapts its toolset based on the detected Modality (e.g., CT, DX, MG) to avoid interface clutter and present only clinically relevant features.

3.1 Volumetric Navigation & Visual Enhancements

- **3D Volume Support:** Uploading a ZIP archive containing a CT sequence automatically activates the Z-Axis Slice Navigator, organizing images temporally/spatially using the ImagePositionPatient tag.
- **Hounsfield Unit (HU) Calibration:** Automatically applies the RescaleSlope (m) and RescaleIntercept (b) to pixel data (PV) for CT images:

$$HU = PV \cdot m + b$$

- **Filters:** Includes spatial filters such as Unsharp Masking for edge enhancement and Median Filtering for noise suppression.
- **3D Volume Projections (MIP & MinIP):** For volumetric CT datasets, the navigator seamlessly switches between Single Slice view, Maximum Intensity Projection (MIP), and Minimum Intensity Projection (MinIP) across Axial, Coronal, and Sagittal planes. MIP effectively isolates high-attenuation structures (e.g., bone, contrast-enhanced vessels, phantom markers), while MinIP is utilized for low-attenuation paths like airways.

3.2 Universal Measurement Tools

These tools are available across all imaging modalities to optimize visual inspection and quantitative assessment.

- **Quick Adjustments & Spatial Filtering:** Real-time optimization tools specifically configured for clinical physics evaluation:
 - **Brightness & Contrast:** Manual sliders for direct intensity mapping independent of DICOM window tags.
 - **Gamma Correction (γ):** Non-linear adjustment to enhance low-intensity structural details without saturating high-intensity regions.
 - **Sharpness Enhancement:** A progressive slider adjusting spatial edge definition.

- **Advanced Spatial Filters:** Selectable kernel operations including Unsharp Masking (Pro Edge), Median Filter (Noise Reduction), Smoothing/Blur, and Histogram Equalization.
- **Multi-ROI Analysis:** Deploy up to 5 regions of interest (Center, Top, Bottom, Left, Right) using circular or rectangular geometries. Calculates local Mean, Standard Deviation (Noise), Min, and Max values, while simultaneously computing the absolute physical area (mm^2) of the selected region using the DICOM PixelSpacing attribute.
- **Distance Ruler & Auto-Calibration:** Extracts spatial resolution from the PixelSpacing tag. Millimeter conversions are calculated automatically.
- **Live Pixel / HU Probe:** A precision crosshair (5 px) that displays real-time spatial coordinates (X, Y) and intensity values without obstructing the diagnostic view.
- **Goniometer (Angle Tool):** Measures anatomical or geometric angles (θ°). By placing a vertex (V) and two arm points (A, B), the internal angle is derived via the dot product of vectors $\vec{u}$ and $\vec{v}$:

$$\theta = \arccos\left(\frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|}\right)$$

- **Interactive Zoom Loupe:** Generates an isolated $2 \times$ or $4 \times$ magnified sub-plot for inspecting microcalcifications or evaluating phantom high-contrast spatial resolution line pairs.
- **Geometric Distortion & Grid Overlay:** Projects a dynamic $1 \text{ cm} \times 1 \text{ cm}$ geometric grid directly over the active image by computing coordinates from the DICOM PixelSpacing tag. This tool provides an immediate visual reference for inspecting spatial distortion (e.g., pincushion or barrel distortion) and verifying laser/phantom alignment.

3.3 Spatial Resolution Analysis (ESF & MTF)

Essential for evaluating system sharpness. By drawing a line profile across a sharp edge (e.g., a phantom insert or collimator blade), the system computes the spatial frequency response.

1. **Edge Spread Function (ESF):** The raw pixel intensity profile across the boundary.
2. **Line Spread Function (LSF):** The first derivative of the ESF.

3. **Modulation Transfer Function (MTF):** Derived from the Fast Fourier Transform (FFT) of the normalized LSF. The system automatically extracts:
 - MTF_{50} : The frequency (cycles/mm) at 50% contrast.
 - MTF_{10} : The limiting spatial resolution frequency.

3.4 Dynamic Modality-Specific QC Tests

3.4 Dynamic Modality-Specific QC Tests The toolkit intelligently triggers specialized QC algorithms based on the uploaded file type.

A. Computed Tomography (CT) - Advanced QA Suite Activated exclusively for Modality == "CT". Based on AAPM Report No. 74, TG-66, and TG-233 standards.

- **Water Calibration Check:** Evaluates the central ROI against the strict tolerance of 0 ± 4 HU.
- **Field Uniformity:** Compares the peripheral ROIs to the central ROI.

$$Uniformity \% = 100 \times \left(1 - \frac{|Mean_{periph} - Mean_{center}|}{|Mean_{center}|} \right)$$

- **Noise (SD):** Assesses the standard deviation within the central ROI as a baseline for image noise.
- **CT Linearity & Sensitometry:** Correlates measured Hounsfield Units (HU) to Relative Electron Density (ρ_e) for phantom inserts (e.g., Air, Water, Acrylic, Delrin, Teflon). Performs linear regression to generate a calibration curve compliant with AAPM TG-66.
- **Slice Thickness Verification (FWHM):** Analyzes the spatial intensity profile across a phantom ramp or wire. Calculates the Full Width at Half Maximum (FWHM) and automatically applies the necessary trigonometric correction (e.g., $\tan(23^\circ)$) to output the true z-axis slice thickness.
- **Noise Power Spectrum (NPS):** Computes the 2D Fast Fourier Transform (FFT) of a homogeneous central Square ROI. Evaluates the spatial frequency distribution and noise texture to benchmark reconstruction algorithms, adhering to AAPM TG-233 guidelines.

B. Projection Radiography (DX / CR) - Collimator Alignment QC

- **Light/Radiation Field Alignment:** Assesses the geometric coincidence between the visual light field and the exposed X-ray radiation field.

- **Workflow:** The user inputs the Source-to-Image Distance (SID) and physical collimator light field settings. By mapping the edges of the exposed radiation field on the image, the tool calculates the geometric deviations (ΔX , ΔY) in millimeters.
- **Compliance:** Automatically assigns a Pass/Fail status based on the standard international tolerance limit of $\leq 2\%$ of the SID.

C. Flat-Field Uniformity & Bad Pixel Detector (DX / CR / MG) Activated for projection radiography and mammography.

- Detects localized artifacts and non-uniformities across the digital flat-panel detector (FPD).
- Evaluates pixels against a statistical $\pm 3\sigma$ threshold derived from the global mean.
- Generates a "Defective Pixel Overlay" and calculates overall detector homogeneity.

D. Figure of Merit (FOM) Calculator (Mammography - MG) Activated exclusively for Modality == "MG". Adheres to EUREF guidelines for protocol optimization (evaluating Target/Filter combinations and tube voltage).

- Integrates contrast-to-noise ratio (CNR) and Mean Glandular Dose (MGD).

$$FOM = \frac{CNR^2}{MGD}$$

- **Workflow:** The user sets the Target ROI (e.g., an insert detail) and a Background ROI. The application calculates the FOM directly, enabling rapid objective comparisons between exposure techniques.

4. Module 3: Batch DRL & Dataset CSV Report Generator

This batch processing module automates the extraction of dosimetric variables required to establish and audit Diagnostic Reference Levels (DRLs) per international protocols (e.g., ICRP, EFOMP).

Users pre-select the modality category before uploading a ZIP archive containing multiple studies.

- **Radiography (DX/CR):**
 - Extracts: kVp, mA, Exposure Time (s), mAs, Source-to-Detector Distance (SID).

- Dosimetry: Total DAP / KAP ($Gy \cdot cm^2$), Entrance Surface Dose (mGy).
- Geometry: Physical field size at the detector plane ($mm \times mm$), dynamically derived from FieldOfViewDimensions or ImagerPixelSpacing.
- **Mammography (MG):**
 - Extracts: kVp, mAs, Target/Filter combinations (e.g., W/Rh), Laterality, Projection (CC/MLO).
 - Specific Dosimetry: **MGD (Mean Glandular Dose)**, ESAK, Compressed Breast Thickness (mm), Compression Force (N).
- **Computed Tomography (CT Volumes):**
 - Automatically groups independent slices into discrete patient/series volumes to prevent duplicate entries.
 - Extracts: kVp, Acquisition Mode (Helical/Sequential), Slice Thickness.
 - Dosimetry: **$CTDI_{vol}$ (mGy), Scan DLP ($mGy \cdot cm$), Total DLP.**
 - Auto-calculations: Computes estimated **Scan Length (cm)** using $L = DLP/CTDI_{vol}$ and assigns dose parameters explicitly to **Head (16 cm)** or **Body (32 cm)** phantoms based on anatomical header inspection.
- **Fluoroscopy & Interventional (XA / RF / RDSR):**
 - Parses standardized Radiation Dose Structured Reports (RDSR) to directly extract cumulative metrics, bypassing the need for optical character recognition on "burned-in" image dose summaries.
 - **Extracts:** Fluoroscopy Time (s), Cumulative Air Kerma (mGy), and total DAP/KAP ($Gy \cdot cm^2$).

5. Module 4: Protocol Auditor & Comparator

This module functions as an automated clinical audit tool, designed to detect unauthorized parameter deviations from a department's established "Gold Standard" protocol.

- **Workflow:** The physicist uploads a reference protocol DICOM alongside a clinical scan from the daily routine.
- **Parameter Cross-Checking:** The system automatically evaluates core geometric and dosimetric metadata, including Modality, Body Part Examined, Tube Voltage (kVp), Tube Current (mA), Exposure Time, Slice Thickness, CTDIvol, Focal Spots, Convolution Kernel, and Pixel Spacing.
- **Compliance Scoring:** Generates a comparative table with color-coded status indicators (MATCH / DEVIATION) and calculates an overall **Protocol Compliance Score (%)** to ensure strict clinical governance and safety protocol adherence.

6. References & Standards

1. **AAPM Report No. 74:** *Quality Control in Diagnostic Radiology*, American Association of Physicists in Medicine.
2. **AAPM Report No. 233:** *Performance Evaluation of Computed Tomography Systems*.
3. **EUREF:** *European Guidelines for Quality Assurance in Breast Cancer Screening and Diagnosis*, 4th Edition.
4. **DICOM Standard (PS3.15 Annex E):** *Basic Application Level Confidentiality Profile*, NEMA.
5. **ICRP Publication 135:** *Diagnostic Reference Levels in Medical Imaging*, International Commission on Radiological Protection.